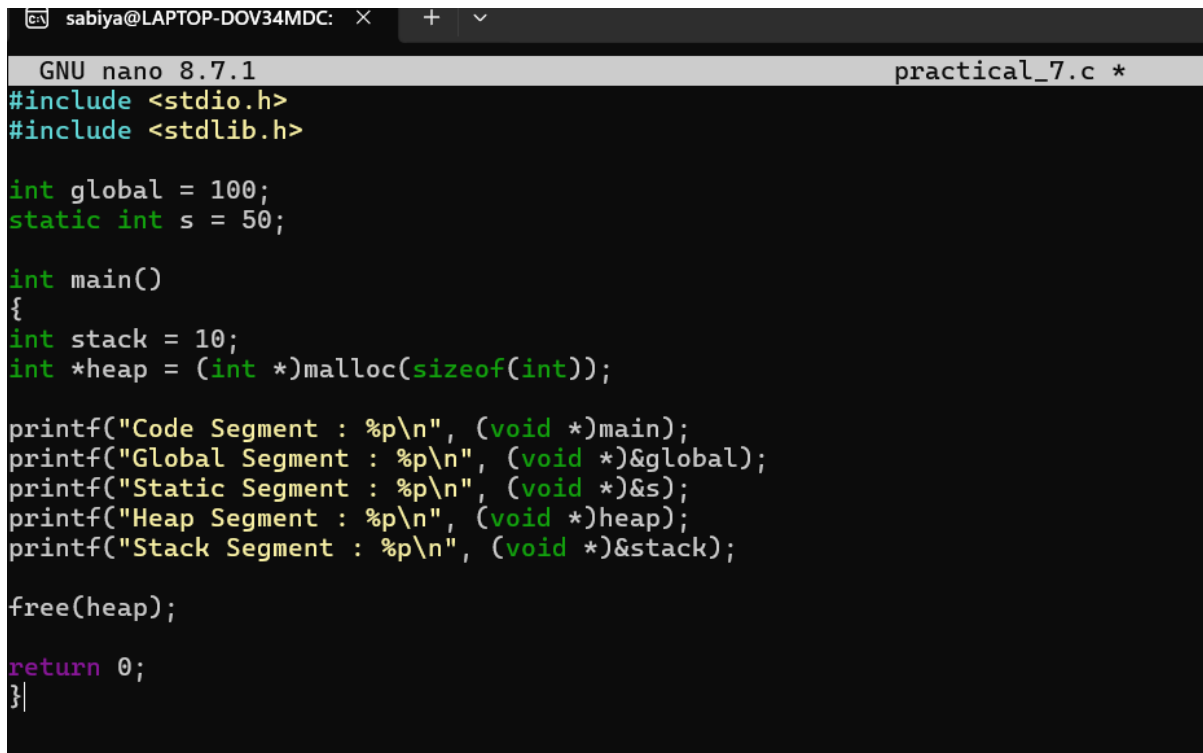


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Practical(7&8)

7.



```
sabiya@LAPTOP-DOV34MDC: × + ▾
GNU nano 8.7.1 practical_7.c *
#include <stdio.h>
#include <stdlib.h>

int global = 100;
static int s = 50;

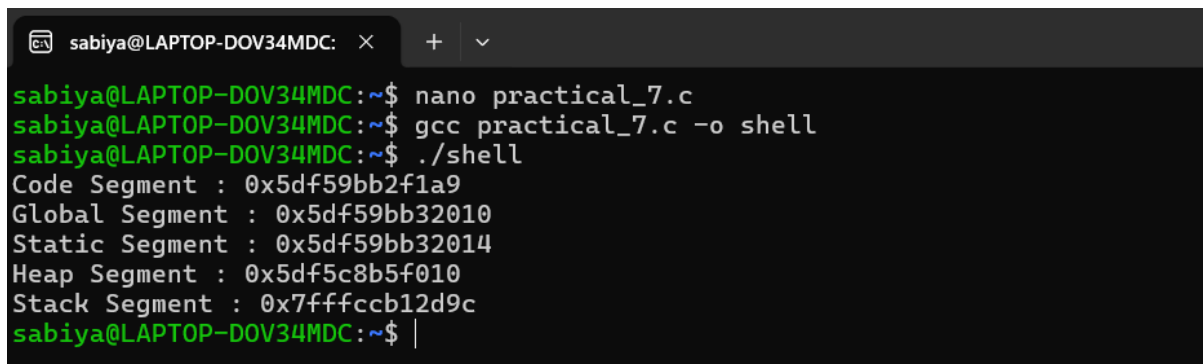
int main()
{
int stack = 10;
int *heap = (int *)malloc(sizeof(int));

printf("Code Segment : %p\n", (void *)main);
printf("Global Segment : %p\n", (void *)&global);
printf("Static Segment : %p\n", (void *)&s);
printf("Heap Segment : %p\n", (void *)heap);
printf("Stack Segment : %p\n", (void *)&stack);

free(heap);

return 0;
}|
```

OUTPUT SCREENSHOT:



```
sabiya@LAPTOP-DOV34MDC: × + ▾
sabiya@LAPTOP-DOV34MDC:~$ nano practical_7.c
sabiya@LAPTOP-DOV34MDC:~$ gcc practical_7.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
Code Segment : 0x5df59bb2f1a9
Global Segment : 0x5df59bb32010
Static Segment : 0x5df59bb32014
Heap Segment : 0x5df5c8b5f010
Stack Segment : 0x7ffffccb12d9c
sabiya@LAPTOP-DOV34MDC:~$ |
```

8(a) CODE:

```
sabiya@LAPTOP-DOV34MDC: × + ▾
GNU nano 8.7.1 practical_8a.c *
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *a, *b;

    a = (int *)malloc(5 * sizeof(int));

    b = (int *)calloc(5, sizeof(int));

    a = (int *)realloc(a, 10 * sizeof(int));

    free(a);
    free(b);

    return 0;
}|
```

OUTPUT SCREENSHOT:

```
Stack Segment : 0x7ffffb12d9c
sabiya@LAPTOP-DOV34MDC:~$ nano practical_8a.c
sabiya@LAPTOP-DOV34MDC:~$ gcc practical_8a.c -o shell
sabiya@LAPTOP-DOV34MDC:~$ ./shell
```

```
sabiya@LAPTOP-DOV34MDC:~$ mkdir ~/Practical
sabiya@LAPTOP-DOV34MDC:~$ cd ~/Practical
sabiya@LAPTOP-DOV34MDC:~/Practical$ pwd
/home/sabiya/Practical
sabiya@LAPTOP-DOV34MDC:~/Practical$ mkdir practical8a
cd practical8a
sabiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ pwd
/home/sabiya/Practical/practical8a
sabiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ nano practical_8a.c
sabiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ gcc practical_8a.c -o shell
sabiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ ./shell
sabiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ valgrind --leak-check=full ./shell
Command 'valgrind' not found, but can be installed with:
sudo snap install valgrind # version 3.26.0, or
sudo apt install valgrind # version 1:3.26.0-0ubuntu1
See 'snap info valgrind' for additional versions.
sabiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ sudo apt install valgrind
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Upgrading:
  libc-bin libc-dev-bin libc-gconv-modules-extra libc6 libc6-dev locales
Installing:
  valgrind
```

```

abiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
72 packages can be upgraded. Run 'apt list --upgradable' to see them.
abiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ valgrind --leak-check=full ./shell
==2171== Memcheck, a memory error detector
==2171== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2171== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2171== Command: ./shell
==2171==
==2171==
==2171== HEAP SUMMARY:
==2171==     in use at exit: 0 bytes in 0 blocks
==2171==   total heap usage: 3 allocs, 3 frees, 80 bytes allocated
==2171==
==2171== All heap blocks were freed -- no leaks are possible
==2171==
==2171== For lists of detected and suppressed errors, rerun with: -s
==2171== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
abiya@LAPTOP-DOV34MDC:~/Practical/practical8a$

```

8(b) CODE:

```

GNU nano 8.7.1 practical_8b.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int x = 100;

    if (fork() == 0)
    {
        printf("Child Before = %d\n", x);

        x = 500;

        printf("Child After = %d\n", x);
    }
    else
    {
        sleep(2);
        printf("Parent = %d\n", x);
    }

    return 0;
}

```

OUTPUTSCREENSHOT:

```
sabiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ nano practical_8b.c
sabiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ gcc practical_8b.c -o shell
sabiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ ./shell
Child Before = 100
Child After = 500
Parent = 100
sabiya@LAPTOP-DOV34MDC:~/Practical/practical8a$ |
```